Name _____ Period _____ Date _____

2-1 Lesson Quiz: Vertex Form of a Quadratic Function

3. What is the equation written in vertex form of a parabola with a vertex of $(9, -1)$ that passes through $(7, 7)$?

Ⓐ $y = 2(x - 9)^2 - 1$

Ⓑ $y = 2(x + 1)^2 - 9$

Ⓒ $y = 2(x - 7)^2 + 7$

Ⓓ $y = 7(x - 9)^2 - 1$

Show how you found a.

$y = a(x - ____)^2 + ____$

4. Which of the following is an equation in the form $y = ax^2 + bx + c$ of the parabola shown in the graph?

Ⓐ $y = -5x^2 - 20x - 65$

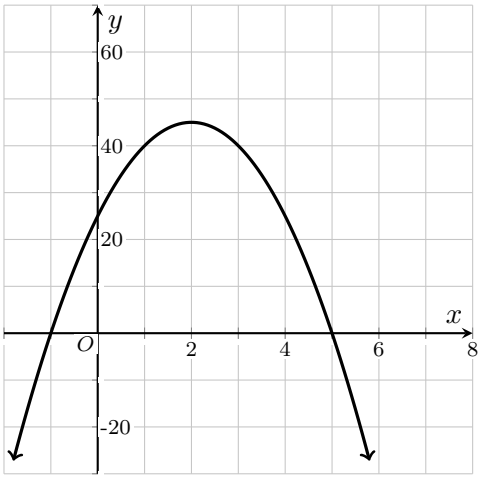
Ⓑ $y = -5x^2 - 20x - 20$

Ⓒ $y = -x^2 - 27x - 90$

Ⓓ $y = -5x^2 + 20x + 25$

From the graph: vertex (_____, _____) y-intercept _____

Show how you found a:



5. Function g is a transformation of the parent function $f(x) = x^2$. The graph of g is a translation right 3 units and down 5 units of the graph of f. What is the equation of function g written in the form $y = ax^2 + bx + c$?

Ⓐ $y = x^2 + 6x + 4$ Ⓑ $y = x^2 - 6x + 4$ Ⓒ $y = x^2 + 10x + 22$ Ⓓ $y = x^2 - 6x + 14$

Show your work. Vertex form first: $g(x) =$ _____ Then expand:

Scoring: Question 4 and Question 5 are 10 points each. Question 3 is a bonus of up to 5 points. Total out of 20. (Questions 1 and 2 of the Savvas quiz are not on this form.)

2-1 Lesson Quiz, short form: answer key

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3. A, $y = 2(x-9)^2 - 1$**.** $y = a(x-9)^2 - 1$; substitute $(7, 7)$: $7 = a(7-9)^2 - 1$, $8 = 4a$, $a = 2$. Distractor C puts the point where the vertex goes; D uses the point's y for a. DOK 2, skill F60. Taught by Example 4 steps 1–3 (Example 3 is the same substitution; cut). Topic 2 test item 1 (the golf ball) is this move in context. **Bonus 5:** 5 for A with the substitution shown; 3 for A with no work.

4. D, $y = -5x^2+20x+25$**.** From the graph the vertex is $(2, 45)$ and the y-intercept is 25. $25 = a(0-2)^2+45$, $-20 = 4a$, $a = -5$. $y = -5(x-2)^2 + 45 = -5(x^2 - 4x + 4) + 45 = -5x^2 + 20x + 25$. Check: $c = 25$ is the y-intercept; the zeros $x = -1$ and $x = 5$ match the graph. DOK 2, skill F16. Taught by Example 4 (Try It 4). **10 points:** 6 for D, 2 for the vertex and y-intercept read correctly, 2 for $a = -5$ shown.

5. B, $y = x^2 - 6x + 4$**.** Right 3 and down 5 move the vertex to $(3, -5)$, and a translation keeps $a = 1$: $g(x) = (x-3)^2 - 5 = x^2 - 6x + 9 - 5 = x^2 - 6x + 4$. Distractor A uses $(x+3)^2$ (left 3); C uses $(x+5)^2 - 3$ (the 3 and 5 swapped); D uses $(x-3)^2 + 5$ (up 5). DOK 2, skills F16, F68. Taught by Example 5 (Try It 5a). Topic 2 test item 2 is this move after a reflection in the x-axis. **10 points:** 6 for B, 2 for the vertex form $g(x) = (x-3)^2 - 5$, 2 for the expansion shown.

Items left off this form (available for a retake or Form B): Savvas item 1, select all true statements about $g(x) = (x+2)^2 + 4$ as a transformation of $f(x) = x^2$ (answer B, C; DOK 2, F60 and F68; Example 1, cut); Savvas item 2, $h(x) = -\frac{1}{2}(x-4)^2 + 3$ is a bird's height, which point is the greatest height (answer A, $(4, 3)$; DOK 2, F60; Example 2, cut).

IXL replacement: SmartScore 70 or above on URV and KQL replaces this quiz with 20/20.